Improving $R(3,k)$ in just two bites

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Ramsey's Theorem

Theorem (Ramsey '29)

For all natural numbers ℓ and k, and for large enough n, every graph on n vertices contains either a clique on ℓ vertices or an independent set on k vertices.

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The smallest such n is called the Ramsey number $R(\ell, k)$.

Bounds for Ramsey numbers

	3	4	5	6	7	8	9	10	11
3	6	9	14	18	23	28	36	40-41	47-50
4		18	25	36-40	49-58	59-79	73-105	92-135	102-170
5			43-46	59-85	80-133	101-193	133-282	149-381	183-511
6				102-147	115-270	134-423	183-651	204-944	*
7					205-492	*	*	*	*
8						282-1518	*	*	*

$R(5,5) \leq 46$: Angeltveit, McKay '24
$R(6,6) \leq 147$: Norin, Volec, Turcotte (unpublished)

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$$2^{k/2+o(1)} \leq R(k,k) \leq (4-c)^k$$

Lower: Erdős '47

$c = 0$: Erdős, Szekeres '35

$c > 0$: Campos, Griffiths, Morris, Sahasrabudhe '23

$c \approx 0.2$: Gupta, Ndiaye, Norin, Wei '24

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$$(c + o(1))\frac{k^2}{\log k} \leq R(3,k) \leq (C + o(1))\frac{k^2}{\log k}$$

$C < \infty$: Ajtai, Komlós and Szemeŕedi '81

$C = 1$: Shearer '83

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$$(c + o(1))\frac{k^2}{\log k} \leq R(3, k) \leq (1 + o(1))\frac{k^2}{\log k}$$

$c > 0$: Kim '95

$c = \frac{1}{4}$: Bohman, Keevash '13, Fiz Pontiveros, Griffiths, Morris '13

$c = \frac{1}{3}$: Campos, Jenssen, Michelen, Sahasrabudhe '25

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$$c\frac{k^3}{\log^4 k} \leq R(4,k) \leq (1+o(1))\frac{k^3}{\log^2 k}$$

Upper: Ajtai, Komlós, Szemerédi '80; Li, Rousseau, Zang '01

Lower: Mattheus, Verstraëte '24

Bounds for Ramsey numbers

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$$c\frac{k^{s-1}}{\log^{2s-4} k} \leq R(s,k) \leq (1+o(1))\frac{k^{s-1}}{\log^{s-2} k}$$

Upper: Ajtai, Komlós, Szemerédi '80; Li, Rousseau, Zang '01

Lower: Bradač '26

THEOREM (HEFTY, HORN, KING, P. '25)

$$R(3,k) \geq \left(\frac{1}{2} + o(1)\right)\frac{k^2}{\log k}.$$

Dual formulation for $R(3, k)$

$R(3, k) = n$:

- Minimize n such that every triangle-free graph on n vertices has $\alpha(G) \geq k$.

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Heuristic:

- Random graphs are good at having low independence number for a given edge density.
- In triangle-free graphs, neighborhoods of vertices are independent.

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Heuristic:

- Random graphs are good at having low independence number for a given edge density.
- In triangle-free graphs, neighborhoods of vertices are independent.
- We want to match the edge density to the independence number of a random graph.

A Conjecture

Conjecture (Campos, Jenssen, Michelen, Sahasrabudhe '25)

$$R(3,k) = \left(\frac{1}{2} + o(1)\right)\frac{k^2}{\log k}.$$

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Equivalent to the lower bound:

Conjecture

There exist triangle-free graphs on n vertices with independence number

$$\alpha(G) = (1 + o(1))\sqrt{n \log n}.$$

The edge deletion threshold

For a fixed graph H, let p_H be the probability that the expected number of copies of H in a random graph $G(n, p_H)$ equals the expected number of edges $\frac{1}{2}p_H n^2$.

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Edge deletion method:
Start with a random graph G_R in $G(n, p)$ for some some $p \ll p_H$, and delete all copies of H by deleting one edge in each copy. Then, heuristically the resulting graph G still looks similar enough to the random graph that the independence number stayed the same:

$$\alpha(G) = (1 + o(1))2\frac{\log(pn)}{p}.$$

The edge deletion bound for K_3

Theorem (Spencer '77)

$$R(3,k) \geq (c + o(1))\,\frac{k^2}{\log^2 k}.$$

Note for $G = G(n, p_{K_3})$:

$$p_{K_3} = \frac{1}{\sqrt{n}}, \quad \bar{d}(G) = \sqrt{n}, \quad \alpha(G) = (1 + o(1))\sqrt{n \log^2 n}.$$

Beating the deletion threshold – Nibble

Idea:

- Instead of going to G in one step, start with a much smaller density p_1, and delete the few edges needed to get rid of all copies of H.
- Continue with a small dose of density p_2 random edges, and again delete all copies of H, but this time only choose the deleted edges from the new edges.
- Continue many steps of this process with homeopathic doses of edges, until almost no further edges can be added.
- This process more carefully squeezes edges into the random graph created this way, ending up with a higher density if the parameters are chosen carefully.
- If one can preserve certain pseudo-random properties through the process, the resultimg graph has independence number similar to a random graph of the same density higher than p_H.

Kim's result

Theorem (Kim '95)

$$R(3,k) \geq (c+o(1))\,\frac{k^2}{\log k} \qquad \textit{for } c \approx \tfrac{1}{160}.$$

For $G = G(n,p)$ of the ending density:

$$p = c_1\sqrt{\frac{\log n}{n}}, \quad \bar{d}(G) = c_1\sqrt{n\log n}, \quad \alpha(G) = c_2\sqrt{n\log n}.$$

Beating the deletion threshold
the H-free process

- Pick one random edge at a time. Add it to the graph if it does not complete a copy of H.
- Stop when no more edges can be added (or when the analysis gets too complicated).
- This is the most incremental nibble imaginable.

Beating the deletion threshold the H-free process

- Pick one random edge at a time. Add it to the graph if it does not complete a copy of H.
- Stop when no more edges can be added (or when the analysis gets too complicated).
- This is the most incremental nibble imaginable.
- This is typically more difficult to analyze, but may end with higher density.
- Many best known lower bounds for $r(H, K_k)$ are proven this way.

$R(3,k)$ from the triangle-free process

Theorem (Bohman, Keevash '13, Fiz Pontiveros, Griffiths, Morris '13)

$$R(3,k) \geq \left(\frac{1}{4} + o(1)\right) \frac{k^2}{\log k}.$$

Beating the next barrier – Blow-Ups

Theorem (Campos, Jenssen, Michelen, Sahasrabudhe '25)

$$R(3,k) \geq \left(\frac{1}{3} + o(1)\right) \frac{k^2}{\log k}.$$

Beating the next barrier – Blow-Ups

Theorem (Campos, Jenssen, Michelen, Sahasrabudhe '25)

$$R(3,k) \geq \left(\frac{1}{3} + o(1)\right) \frac{k^2}{\log k}.$$

- Start with a smaller random graph G_R in $G(N, p)$ for some $p < p_H$. Delete all copies of H by deleting few edges.
- Take an r-blowup of G_R to get a graph with $n = Nr$ vertices. Choose p and r such that this graph has density greater than p_H.
- Finish with a nibble/H-free process.

More details:

- $r = \log^2 n$, $p = \beta\sqrt{\frac{\log n}{n}} = (\beta + o(1))\sqrt{\frac{1}{N \log N}}$
- Note that $p_{K_3} = \sqrt{\frac{1}{n}}$
- $1 - o(1)$ of the edges in the blow-up are open.
- While the blow-up has a few very large independent sets, the additional edges bring the graph back to pseudo-randomness.

Doubling up on Blow-Ups

Theorem (Hefty, Horn, King, P. '25)

$$R(3,k) \geq \left(\frac{1}{2} + o(1)\right)\frac{k^2}{\log k}.$$

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Idea:

- Take red and blue copies of the Blow-Up of CJMS and randomly glue them together/take the union.
- In every resulting triangle, delete the edge of the minority color.
- Because almost all pairs are open, almost all edges survive.
- The random gluing creates enough randomness that the resulting graph is pseudo-random, and the independence bounds from random graphs hold.

Doubling up on Blow-Ups

Theorem (Hefty, Horn, King, P. '25)

$$R(3,k) \geq \left(\frac{1}{2} + o(1)\right)\frac{k^2}{\log k}.$$

Highlights:

- No nibble required. The most advanced tool is McDiarmid's Inequality.
- For $G = G(n,p)$ of the resulting density:

$$\bar{d}(G) = (1+o(1))\sqrt{n \log n}, \quad \alpha(G) = (1+o(1))\sqrt{n \log n}.$$

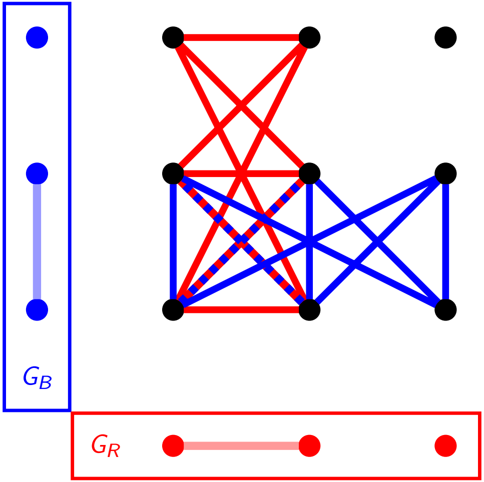
More details

Start with a red and blue random graph G_R and G_B in $G(N, p)$, where

$$p = \frac{1}{2}\sqrt{\frac{\log n}{n}}, \quad r = \log^2 n, \quad rN = n.$$

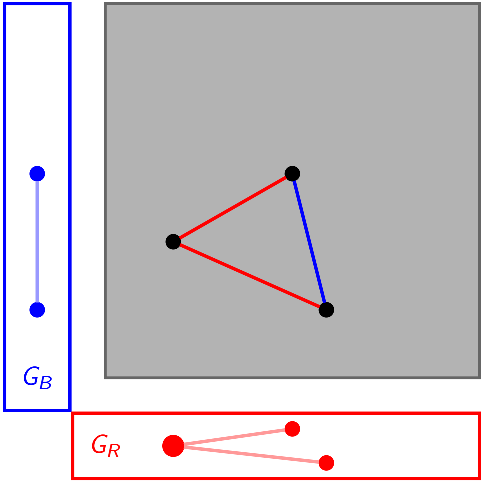
More details

Take the co-normal/OR-product of G_R and G_B:



More details

Delete the last edge in every monochromatic triangle, and the minority edge in every 2-colored triangle.



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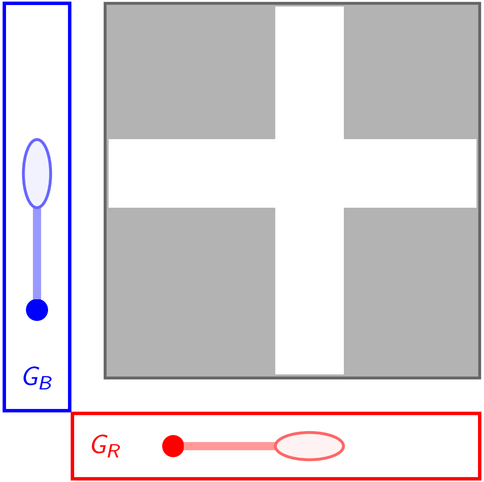

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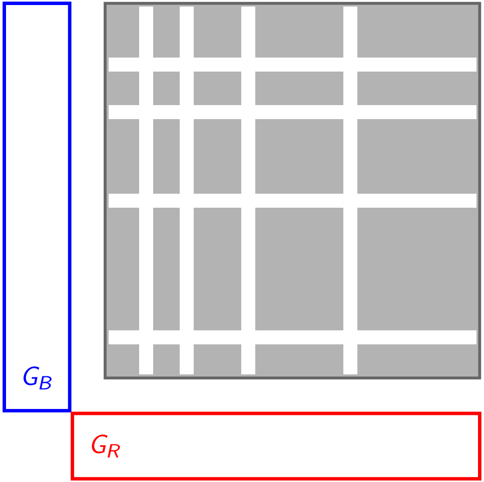
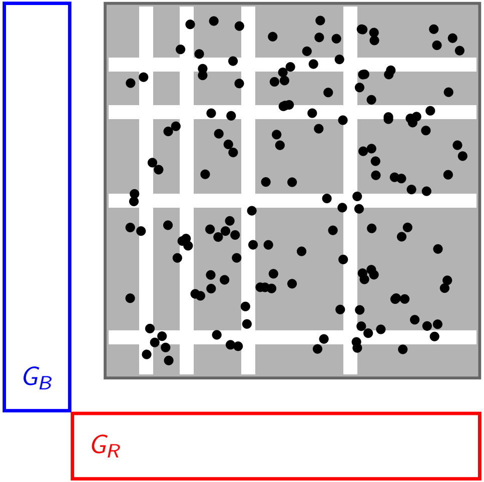


More details

Take a random induced subgraph of size n.



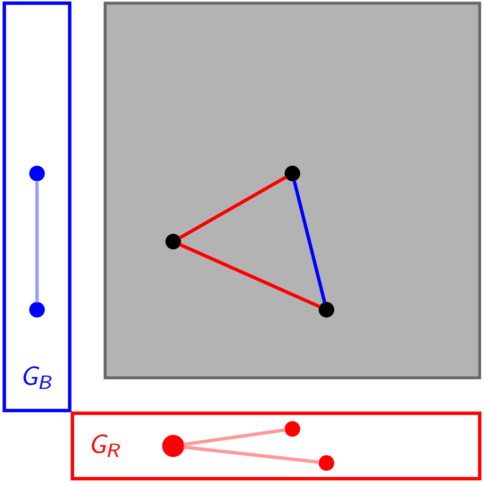
Key Observation

- We expect

$$(2p)^3 n^3 = (\log n)^{\frac{3}{2}} n^{\frac{3}{2}} \qquad \text{triangles,}$$

$$2p\binom{n}{2} = \frac{1}{2}(\log n)^{\frac{1}{2}} n^{\frac{3}{2}} \qquad \text{edges.}$$

Key Observation



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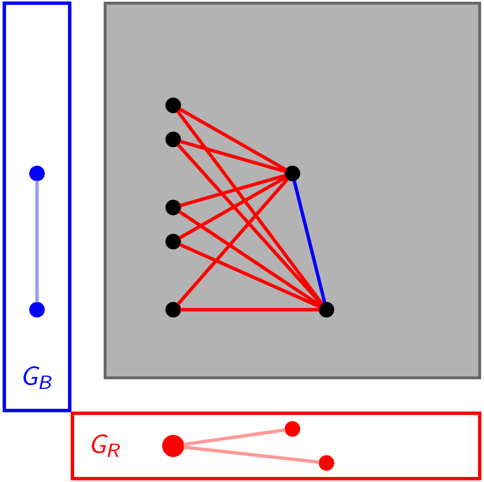

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\begin{aligned}
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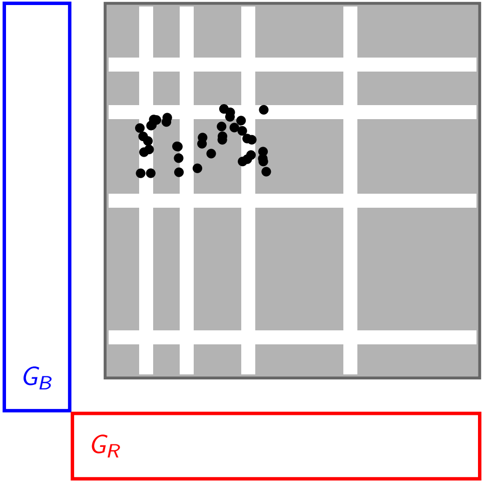
2p\binom{n}{2} &= \frac{1}{2}(\log n)^{\frac{1}{2}} n^{\frac{3}{2}} && \text{edges.}
\end{aligned}
$$

- But: every closed/deleted edge is in about $\log^2 n$ triangles, so we are deleting only $o(1)$ of the edges.

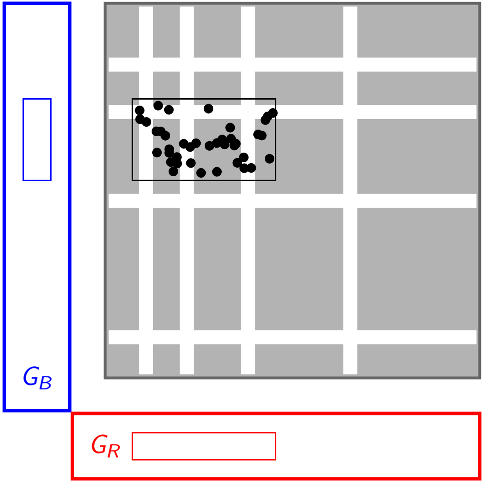
More details

Look at a k-set.



More details

Look at a k-set. We need to look at the projections.



Theorem (Hefty, Horn, King, P. '25)

$$\left(\frac{1}{2} - o(1)\right)\frac{k^2}{\log k} \leq R\left(S_4^{(3)}, S_k^{(3)}\right) \leq (1+o(1))\frac{k^2}{\log k}.$$

This hypergraph Ramsey problem is equivalent to no vertex having a red triangle or a blue K_k in its link graph, and we can use our proof plus 5% extra effort.

Odd Hadwiger Conjecture is false

Conjecture (Odd Hadwiger Conjecture)

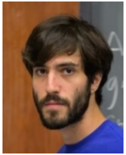
Let G be a graph. If $\chi(G) \geq t$, then G has K_t as an odd minor. Equivalently, if G does not have K_t as an odd minor, then $\chi(G) < t$.

Odd complete Minor: each branch set is a tree with a proper 2-coloring, and every pair of branch sets s connected by a monochromatic edge.

Theorem (Kühn, Sauermann, Steiner, Wigderson '25)

The odd Hadwiger conjecture is false.



Odd Cycles vs Independent Sets

Theorem (Campos, Jenssen, Michelen, P., Sahasrabudhe '25)

For all odd $\ell \geq 5$, we have that

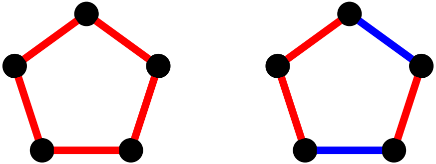
$$r(C_\ell, K_k) \geq k^{1+\frac{1}{\ell-2}+\varepsilon_\ell+o(1)},$$

where $\varepsilon_\ell = ((\ell-2)(2\ell-5))^{-1}$.

New for all $\ell \geq 9$

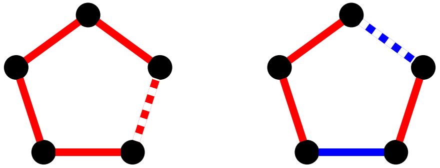
Odd Cycles vs Independent Sets

- Choose an appropriate r with $n = N \cdot r$. Same blow-up construction and product.
- Order the edges in each color.



Odd Cycles vs Independent Sets

- Choose an appropriate r with $n = N \cdot r$. Same blow-up construction and product.
- Order the edges in each color.
- From each monochromatic cycle in the product, delete the largest edge.
- From each remaining cycle in the product, delete the largest edge of the minority color.



Odd Cycles vs Independent Sets

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- Order the edges in each color.
- From each monochromatic cycle in the product, delete the largest edge.
- From each remaining cycle in the product, delete the largest edge of the minority color.
- Root each such cycle in a vertex with two remaining edges of the majority color.